Ayush Agarwal

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Education

- **VIT Bhopal University** — Bhopal, India
  Bachelor of Technology - Computer Science and Engineering; CGPA: 8.52/10 — *Sep 2023 – Ongoing*
  Courses: *Data Structures, Algorithms, Machine Learning, Deep Learning, NLP, Databases, Cloud Computing*

Skills Summary

- **Programming**: Python
- **Backend**: FastAPI, REST APIs, WebSockets, Async Programming, SQLAlchemy
- **AI/ML**: LLMs, RAG, NLP, Prompt Engineering, Multi-Agent Systems, LangChain, MCP, Tool Calling
- **Databases**: SQLite (FTS5), PostgreSQL, Firestore, Azure Cosmos DB
- **Cloud**: Google Cloud Run, Cloud Build, Firebase Hosting, Microsoft Azure, Docker
- **Retrieval**: FAISS, BM25, Vector Search, PageRank, Reciprocal Rank Fusion
- **Libraries**: Scikit-learn, Pandas, NumPy, Streamlit

Experience

- **Docu3C Technologies** — Remote (Seattle, Washington, USA)
  AI Software Engineer Intern — *Aug 2025 – Jun 2026*
  - **Backend Engineering**: Developed asynchronous FastAPI services using REST APIs, Server-Sent Events (SSE), and WebSockets to support concurrent AI inference, streaming responses, and scalable document processing workflows.
  - **Agentic AI Systems**: Designed and implemented multi-agent AI workflows using Model Context Protocol (MCP) with tool-calling capabilities, reducing manual financial document review by 60% across production document intelligence pipelines.
  - **LLM Engineering**: Built production-grade LLM pipelines with structured outputs using Pydantic and Instructor, implementing prompt engineering, validation, and automated extraction across multiple financial document workflows.
  - **Full-Stack Development**: Developed internal AI applications using React and Streamlit, integrated backend APIs, contributed to CRM development and ServiceNow agentic workflows, and deployed solutions on Microsoft Azure infrastructure.
  - **Software Engineering**: Maintained production-quality Python code following modular architecture, PEP8 standards, Git workflows, Ruff linting, modern dependency management (uv), and Agile development practices with Kanban-based sprint planning.

Projects

- **LexAI - Full-Stack Legal Intelligence Platform (Flutter, React, FastAPI, GCP)**
  Web: lexai-3fd1a.web.app GitHub: Repository: Designed and developed a production-grade Legal AI platform indexing **26,274 Supreme Court judgments**. Engineered a Hybrid RAG pipeline combining **BM25 (SQLite FTS5)**, **FAISS dense retrieval**, **PageRank graph ranking**, and **Weighted Reciprocal Rank Fusion** for high-precision precedent retrieval. Built secure REST and WebSocket APIs using FastAPI, optimized concurrent database access with SQLite WAL mode, containerized the backend using Docker, and deployed it on Google Cloud Run with Firebase-hosted web and Flutter mobile clients.
- **PRESCRIPTION - AI Voice Medical Assistant (FastAPI, Whisper, Groq, Docker)**
  Live: Hugging Face Space GitHub: Repository: Developed an AI-powered voice medical assistant that converts doctor consultations into structured digital prescriptions. Integrated Faster-Whisper with LLM-based medical entity extraction, implemented fuzzy medicine matching across **30,000+ Indian medicines** using RapidFuzz, generated printable PDF prescriptions, exposed FastAPI APIs, containerized the application using Docker, and validated functionality through a comprehensive PyTest suite.
- **Legal Knowledge Graph Pipeline (spaCy, OCR, NetworkX)**
  GitHub: Repository: Engineered an end-to-end NLP pipeline that transforms unstructured legal PDFs into canonical knowledge graphs using OCR, spaCy NER, RapidFuzz, and NetworkX. Implemented entity normalization, deterministic ID mapping, relation standardization, and graph construction to support semantic retrieval, downstream analytics, and interactive visualization.

Certifications

- **Microsoft Certified: Azure Data Fundamentals**: Microsoft (Jun 2025)
- **Introduction to Machine Learning**: NPTEL (May 2025)
- **Introduction to Large Language Models**: Google Cloud (Apr 2024)
- **Introduction to Generative AI**: Google Cloud (Apr 2024)
- **Prompt Design in Vertex AI**: Google Cloud (Jul 2024)

Achievements & Activities

- Top 25 team, Mahakumbh Hackathon
- Solved 125+ Data Structures and Algorithms problems on LeetCode
- Event Lead, Linpack Club, VIT Bhopal - Organized 5+ technical workshops with 200+ participants
- Represented university volleyball team (Thunders) at Aavahan sports fest, VIT Bhopal